

defined by the ideal sheaf $\mathcal{I}_X^p(X|Y) = \mathcal{I}_X^p(\Omega_X|Y)$, called the Jacobian subschemes of X over Y . For each point $x \in X$, we set $X_x = \text{Spec}(\mathcal{O}_{X,x})$.

Theorem 5.3. Let $X \rightarrow Y$ be a morphism of essentially finite presentation. Then

(1) For any base change $Y \leftarrow Y'$, the canonical map

$$J_X^p(X'|Y') \cong J_X^p(X|Y) \times_{Y'} Y' \text{ is an isomorphism for all } p \text{ where } X' = X \times_Y Y'.$$

(2) For each point $x \in X$, we have the canonical isomorphism

$$J_{X_x}^p(X_x|Y) \cong J_X^p(X|Y)_x.$$

(3) $J_X^0(X|Y) \supset J_X^1(X|Y) \supset J_X^2(X|Y) \supset \dots$, and $x \notin J_X^p(X|Y)$ for all $p \geq \dim_{\kappa(x)} \Omega_X|Y(x)$.

(4) $x \notin J_X^p(X|Y)$ if and only if there exists an open neighbourhood U of x which admits a closed immersion into a Y -scheme U' such that U' is smooth over Y at x' and $\dim_{x'} U'_{f(x)} = p$ where x' is the image of x under $U \rightarrow U'$.

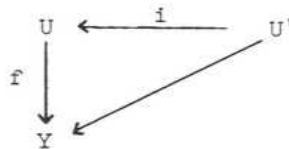
$$\mathcal{I}_{X_1 \times_Y X_2}^r(X_1 \times_Y X_2|Y) = \sum_{p+q=r} \pi_{1-X_1}^* \mathcal{I}_{X_1}^p(X_1|Y) \cdot \pi_{2-X_2}^* \mathcal{I}_{X_2}^q(X_2|Y)$$

where $\pi_i : X_1 \times_Y X_2 \rightarrow X_i$ is the projection.

Proof: (3) follows from 5.1(b) whereas (1) follows from 5.1(a) together with the fact that $P^* \Omega_{X|Y} \cong \Omega_{X'|Y'}$ is an isomorphism where $P : X \times_Y Y' \rightarrow X$ is the projection. (2) also follows from 5.1(a) since $\mathcal{O}_{X,x} \otimes_{\mathcal{O}_X} \Omega_X|Y \cong \Omega_{X_x}|Y$ is an isomorphism (5) follows from 5.1(d) and the isomorphism

$$\pi_{1-X_1}^* \Omega_{X_1}|Y \oplus \pi_{2-X_2}^* \Omega_{X_2}|Y \cong \Omega_{X_1 \times_Y X_2}|Y$$

(4) Assume the existence of a diagram



where U' is smooth over Y at $x' = i(x)$ and $\dim_{x'} U'_f(x) = p$. Then $\dim_{\kappa(x')} \Omega_{U'|Y}(x') = p$ and $i^* \Omega_{U'|Y} \rightarrow \Omega_{U|Y} \rightarrow 0$ is exact, so that $\dim_{\kappa(x)} \Omega_{U|Y}(x) \leq p$ which implies that $x \notin J_U^p(U|Y)$ by 5.3(3). Since $J_U^p(U|Y) = J_X^p(X|Y)|_U$, this means that $x \notin J_X^p(X|Y)$. Now assume conversely that $x \notin J_X^p(X|Y)$.

The question being local, we may assume that X, Y are both affine, say $X = \text{Spec}(B)$, $Y = \text{Spec}(A)$. Choose a presentation over A

$$0 \rightarrow J \rightarrow P \rightarrow B \rightarrow 0$$

where $P = A[x_1, \dots, x_n]$. Then $J/J^2 \xrightarrow{d} B \otimes_P \Omega_{P|A} \rightarrow \Omega_{B|A} \rightarrow 0$ is exact and

$B \otimes_P \Omega_{P|A}$ is B -free of rank n , and therefore

$I_B^p(\Omega_{B|A}) = \text{Im}(\wedge^p J/J^2 \otimes_B (B \otimes_P \wedge^p \Omega_{P|A})^* \rightarrow B)$. Therefore, $x \notin J_X^p(X|Y) \Leftrightarrow$ there exists $f_1, \dots, f_{n-p}$ in J such that $[\det(\frac{f_i}{x_j})](x) \neq 0$ for some choice of $n-p$ variables among $x_1, x_2, \dots, x_n$ there exists $f_1, f_2, \dots, f_{n-p}$ in J such that $\text{Spec}(P/(f_1, \dots, f_{n-p})) \rightarrow \text{Spec}(A)$ is smooth at x' where x' is the image of x under $\text{Spec}(B) \rightarrow \text{Spec}(P/(f_1, \dots, f_{n-p}))$. Thus it suffices to take $U' = \text{Spec}(P/(f_1, \dots, f_{n-p}))$.

The following is a generalization of the usual Jacobian criteria for relative situation.

Theorem 5.4. Let $X \xrightarrow{f} Y$ be a morphism of essentially finite presentation. Assume that there exists a fixed integer $n \geq 0$ with either conditions:

- (1) f is flat and $\dim_{x'} X_{f(x)} = n$ for every point $x \in X$.
- (2) f is a relative complete-intersection with virtual dimension n

(SGA6 VIII 1.9).

Then $x \notin J_X^n(X|Y)$ if and only if f is smooth at x .

Proof: We have, by 5.3(4) that $x \notin J_X^n(X|Y)$ if and only if there exists a diagram

$$\begin{array}{ccc}
 U & \xrightarrow{i} & U' \\
 f \downarrow & \nearrow f' & \\
 Y & &
 \end{array}$$

such that f' is smooth at x' and $\dim_{x'} U'_{f'(x)} = n$, where i is a closed immersion and $x' = i(x)$. Thus it suffices to show that either conditions (1) or (2) entails the isomorphism of i at the point x . Set $A = \mathcal{O}_{Y, f(x)}$, $B' = \mathcal{O}_{U', x'}$, $B = \mathcal{O}_{U, x}$ and let

$$0 \rightarrow I \rightarrow B' \rightarrow B \rightarrow 0$$

be exact. We must show that $I = 0$. Assume (1). Since B is A -flat,

$$0 \rightarrow I/\underline{m}I \rightarrow B'/\underline{m}B' \rightarrow B/\underline{m}B \rightarrow 0$$

is exact where $\underline{m}$ is the maximal ideal of A . By hypothesis, $\dim_x U_{f(x)} = \dim_{x'} U'_{f'(x)}$ so that $\dim B'/\underline{m}B' = \dim B/\underline{m}B$. However $B'/\underline{m}B'$ is a regular local ring and therefore it entails that $I/\underline{m}I = 0$ so that $I = 0$. Now assume (2), and consider the exact sequence

$$I/I^2 \xrightarrow{d} B \otimes_B \Omega_{B'|A} \rightarrow \Omega_{B|A} \rightarrow 0$$

Since X is a relative complete-intersection over Y and $U' \rightarrow Y$ is smooth at x , it follows that $d : I/I^2 \rightarrow B \otimes_B \Omega_{B'|A}$ is injective at x and I/I^2 and $B \otimes_B \Omega_{B'|A}$ are both free at x . Therefore by the definition of virtual dimension of relative complete-intersection we must have that

$$n = \dim_{\kappa(x)} \Omega_{B'|A}(x') - \dim_{\kappa(x)} I/I^2(x) = n - \dim_{\kappa(x)} I/I^2(x)$$

i.e., $I/I^2 = 0$ i.e., $I = 0$.

Definition 5.5. Let $X \xrightarrow{f} Y$ be a morphism of essentially finite presentation. In case when there exists a fixed integer n with the property (1) or (2) in 5.4,

we set $I_X(X|Y) = I_X^n(X|Y)$ and $J_X(X|Y) = J_X^n(X|Y)$.

We can extend this notion to formal schemes in an obvious manner.

Thus let X be a noetherian formal scheme and E a coherent X -module. Then we can define the ideal sheaf $I_X^P(E)$ by setting

$$I_X^P(E)(U) = I_{X(U)}^P(E(U))$$

for every affine open $U \subset X$. If $X = \varprojlim_v X_v$, then $E = \varprojlim_v E_v$ where

$E_v = E \otimes_{O_X} O_{X_v}$ and it follows from the fact that X is noetherian that

$$I_X^P(E) = \varprojlim_v I_{X_v}^P(E_v)$$

Now let S be a noetherian formal scheme and X an S -adic formal scheme of essentially finite type. Let I_X be the ideal sheaf defining the closed immersion $\Delta: X \rightarrow X \times_S X$ (= the fibre-product in the category of formal schemes) i.e.,

$$0 \rightarrow I_X \rightarrow O_X \times_S X \rightarrow O_X \rightarrow 0$$

is exact. We define $\Omega_{X|S} = I_X/I_X^2$. Since X is an S -adic formal scheme of essentially finite type, it follows that $X = \varprojlim_v X_v$ where $X_v = X \times_S S_v$, and hence $\Omega_{X|S} \simeq \varprojlim_v \Omega_{X_v|S_v}$ is a coherent X -module. We note that for any formal S -scheme S' of finite type, we have a canonical isomorphism

$$\Omega_X \times_{S'} S'|_{S'} \simeq O_{S'} \otimes_{O_S} \Omega_{X|S}.$$

Definition 5.6. Let S be a noetherian formal scheme and X an S -adic formal scheme of essentially finite type. We define the ideal sheaf $I_X^P(X|S)$ by

$$I_X^P(X|S) = I_X^P(\Omega_{X|S})$$

and we set $J_X^P(X|S)$ the formal closed subscheme of X defined by $I_X^P(X|S)$.

Since $\Omega_{X|S} = \varprojlim_n \Omega_{X_n|S_n}$ we have

$$I_{\mathcal{X}}^p(\mathcal{X}|S) = \varprojlim_n I_{X_n}^p(X_n|S_n)$$

$$J_{\mathcal{X}}^p(\mathcal{X}|S) = \varinjlim_n J_{X_n}^p(X_n|S_n) .$$

$\mathcal{X}_{(x)}$ is an S -adic formal scheme of essentially finite type and we have a commutative diagram

$$\begin{array}{ccc} \mathcal{X}_{(\hat{x})} & \xrightarrow{i_{(\hat{x})}} & \mathcal{X} \\ & \searrow & \nearrow i_{(x)} \\ & \mathcal{X}_{(x)} & \end{array}$$

We note that if x is an isolated point of $\mathcal{X}$, then $i_{(x)}$, $i_{(\hat{x})}$ are both isomorphisms to the connected component of $\mathcal{X}$ containing the point x .

In particular, if $|\mathcal{X}| = \{x_1, \dots, x_m\}$ is a finite discrete space, then we have a canonical identification $\mathcal{X} = \bigsqcup_{1 \leq i \leq m} \mathcal{X}_{(x_i)}$ (disjoint union).

Corollary 5.7. Let S be a noetherian formal scheme, and $\mathcal{X}$ an S -adic formal scheme of essentially finite type. We then have

(1) $J_{\mathcal{X}}^p(\mathcal{X}|S)$ commutes with base change i.e., $J_{\mathcal{X}'}^p(\mathcal{X}'|S') = J_{\mathcal{X}}^p(\mathcal{X}|S) \times_S S'$ where $\mathcal{X} = \mathcal{X} \times_S S'$.

(2) The connected components of $J_{\mathcal{X}}^p(\mathcal{X}|S)$ is in bijective correspondence with that of $J_{X_0}^p(X_0|S_0)$. For each point $x \in |J_{X_0}^p(X_0|S_0)|$ we have a canonical isomorphism $J_{\mathcal{X}}^p(\mathcal{X}|S)_{(x)} \simeq J_{\mathcal{X}_{(x)}}^p(\mathcal{X}_{(x)}|S)$. In particular if

$$|J_{X_0}^p(X_0|S_0)| = \{x_1, \dots, x_m\}$$

is discrete, then

$$J_{\mathcal{X}}^p(\mathcal{X}|S) = \bigsqcup_{1 \leq i \leq m} J_{\mathcal{X}_{(x_i)}}^p(\mathcal{X}_{(x_i)}|S)$$

(disjoint union).

(3) Let S be an ordinary noetherian scheme and X an S -scheme of essentially finite type. Given a closed subscheme $S_0 \subset S$, let $\hat{S}$ be the formal completion of S along S_0 .

If we set

$\hat{X} = X \times_S \hat{S}$ = the formal completion of X along $X_0 = X \times_S S_0$, then

$J_X^p(\hat{X}|\hat{S}) = J_X^p(X|S) \times_S \hat{S}$ = the formal completion of $J_X^p(X|S)$ along $J_{X_0}^p(X_0|S_0)$.

(4) If there exists a fixed integer $n \geq 0$ with the property 5.4(1) or (2) for $X_v \rightarrow S_v$ for all v , then $x \notin J_X^n(X|S)$ if and only if X is formally smooth over S at x .

Proof: (1) follows from $\Omega_{X'|S'} \simeq \Omega_S' \otimes_{\Omega_S} \Omega_{X|S}$ whereas (2) follows from

$$\Omega_{X(x)} \otimes_{\Omega_{X(x)}} \Omega_{X|S} \simeq \Omega_{X(x)|S}.$$

$$(3) \quad J_X^p(\hat{X}|\hat{S}) = \varinjlim_v J_{X \times_S S_v}^p(X \times_S S_v|S_v) = \varinjlim_v J_X^p(X|S) \times_{S_v} = J_X^p(X|S) \times_S \hat{S}.$$

(4) Since X is an S -adic formal scheme, $X \rightarrow S$ is formally smooth at $x \Leftrightarrow X_v \rightarrow S_v$ is formally smooth at x for all $v \Leftrightarrow x \notin J_{X_v}^n(X_v|S_v)$ for all $v \Leftrightarrow x \notin \varinjlim_v J_{X_v}^n(X_v|S_v) = J_X^n(X|S)$.

Remark 5.8. Let X be an S -adic formal scheme of essentially finite type. Given a point $x \in X$ we also consider the S -adic formal scheme $X_{(\hat{X})} = \varinjlim_v \text{Spec}(\hat{\mathcal{O}}_{X_v, x})$.

We thus have the natural morphisms $X_{(\hat{X})} \rightarrow X_{(x)} \rightarrow X$. Now $\hat{\mathcal{O}}_{X_v, x}$ is in general not essentially finite type over $S_{(y)}$ where y = the image of x under $X \rightarrow S$.

For simplicity, let us set $A_v = \mathcal{O}_{X_v, x}$, $R_v = \mathcal{O}_{S_v, y}$. Now the natural map

$\Omega_{A_v|R_v} \rightarrow \hat{\Omega}_{A_v|R_v}$ induces an isomorphism $\hat{\Omega}_{A_v|R_v} \rightarrow \hat{\Omega}_{A_v|R_v}$, and therefore

$\varinjlim_v \hat{\Omega}_{A_v|R_v} \simeq \varinjlim_v \hat{\Omega}_{A_v|R_v}$ where $\wedge$ stands for the separable completion of a

module with respect to the topology induced from the ring. Therefore if we define

$\hat{\Omega}_{X_{(\hat{X})}|S} = \varinjlim_v \hat{\Omega}_{\hat{\mathcal{O}}_{X_v, x}|\mathcal{O}_{S_v, y}}$, then $\hat{\Omega}_{X_{(\hat{X})}|S}$ is a coherent $X_{(\hat{X})}$ -module. Thus if

we set $J_{\mathcal{X}(\hat{x})}^p(\mathcal{X}(\hat{x})|S)$ to be the formal closed subscheme of $\mathcal{X}(\hat{x})$ defined by the ideal sheaf $I_{\mathcal{X}(\hat{x})}^p(\mathcal{X}(\hat{x})|S) = I_{\mathcal{X}(\hat{x})}^p(\Omega_{\mathcal{X}(\hat{x})}|S)$, then we have

$$J_{\mathcal{X}(x)}^p(\mathcal{X}(x)|S)_{(\hat{x})} \simeq J_{\mathcal{X}(\hat{x})}^p(\mathcal{X}(\hat{x})|S).$$

We note that if x is an isolated point of $J_{\mathcal{X}}^p(\mathcal{X}|S)$, then $J_{\mathcal{X}(\hat{x})}^p(\mathcal{X}(\hat{x})|S) \rightarrow J_{\mathcal{X}(x)}^p(\mathcal{X}(x)|S)$ is an isomorphism.

To consider the "image" of $J_{\mathcal{X}}^p(\mathcal{X}|S)$ under the morphism $\mathcal{X} \xrightarrow{f} S$, one may simply take the formal subscheme of S defined by the ideal sheaves $\text{Ker}(\mathcal{O}_S \rightarrow \mathcal{O}_{\mathcal{X}}/I_{\mathcal{X}}^p(\mathcal{X}|S))$. However, it is more convenient for our purpose to use the Fitting ideal again.

Definition 5.9. Let S be a noetherian formal scheme and $f: \mathcal{X} \rightarrow S$ an S -adic formal scheme of essentially finite type. Assume that $J_{\mathcal{X}}^p(\mathcal{X}|S)$ is proper over S , so that $f_* \mathcal{O}_{J_{\mathcal{X}}^p(\mathcal{X}|S)}$ is a coherent $\mathcal{O}_S$ -module. We then define the ideal sheaf $I_S^p(\mathcal{X}|S)$ of $\mathcal{O}_S$ by

$$I_S^p(\mathcal{X}|S) = I_S^0(f_* \mathcal{O}_{J_{\mathcal{X}}^p(\mathcal{X}|S)})$$

and set $J_S^p(\mathcal{X}|S)$ to be the closed formal subscheme of S defined by the ideal sheaf $I_S^p(\mathcal{X}|S)$. In case when there exists a fixed integer n having the property (1) or (2) in 5.4 for $X_v \rightarrow S_v$ for all v , we set

$$I_{\mathcal{X}}(\mathcal{X}|S) = I_{\mathcal{X}}^n(\mathcal{X}|S), \quad J_{\mathcal{X}}(\mathcal{X}|S) = J_{\mathcal{X}}^n(\mathcal{X}|S)$$

$$I_S(\mathcal{X}|S) = I_S^n(\mathcal{X}|S), \quad J_S(\mathcal{X}|S) = J_S^n(\mathcal{X}|S)$$

Corollary 5.10. Let S be a noetherian formal scheme and $\mathcal{X} \xrightarrow{f} S$ an S -adic formal scheme, essentially of finite type such that $J_{\mathcal{X}}^p(\mathcal{X}|S)$ is proper over S . Then

$$(1) \quad |J_S^p(\mathcal{X}|S)| = \text{the image of } |J_{\mathcal{X}}^p(\mathcal{X}|S)| \text{ under } f: \mathcal{X} \rightarrow S.$$

(2) $J_S^p(\mathcal{X}|S)$ commute with base change i.e., $J_{S'}^p(\mathcal{X}'|S') = J_S^p(\mathcal{X}|S) \times S'$.

(3) If $J_X^p(\mathcal{X}|S) = \bigsqcup_{1 \leq i \leq m} \mathcal{U}^{(i)}$ (disjoint union), then

$$I_S^p(\mathcal{X}|S) = \prod_{1 \leq i \leq m} I_S^{O(f_{*}O_{\mathcal{U}^{(i)}})} . \text{ In particular if}$$

$$|J_{X_0}^p(X_0|S_0)| = \{x_1, \dots, x_m\}$$

is discrete, then

$$I_S^p(\mathcal{X}|S) = \prod_{1 \leq i \leq m} I_S^p(\mathcal{X}_{(x_i)}|S) = \prod_{1 \leq i \leq m} I_S^p(\mathcal{X}_{(x_i)}|S)$$

(4) Let S be an ordinary noetherian scheme and X an S -scheme of essentially finite type. Given a closed subscheme $S_0 \rightarrow S$, let $\hat{S}$ be the formal completion of S along S_0 . If we set $\hat{X} = X \times_S \hat{S}$ = the formal completion of X along $X_0 = X \times_S S_0$, then $J_{\hat{S}}^p(\hat{X}|\hat{S}) = J_S^p(X|S) \times \hat{S}$ = the formal completion of $J_S^p(X|S)$ along $J_{S_0}^p(X_0|S_0)$.

Proof: $\text{Supp } f_* O_{\mathcal{U}} = |J_S^p(\mathcal{X}|S)|$ by 5.1(c), and hence (1) follows. (2) and (4)

are clear from 5.1(a), and (3) follows from 5.1(d), and the isomorphism

$$J_{\mathcal{U}^{(i)}}^p(\mathcal{X}_{(x)}|S) \cong J_{\mathcal{U}^{(i)}}^p(\mathcal{X}_{(x)}|S) \text{ for an isolated point } x \text{ in } J_{\mathcal{U}}^p(\mathcal{X}|S) .$$

Remark 5.11. Our definition of Jacobian ideal, adopted from the classical one over base field, commutes with base change and gives a smoothness criteria under a mild condition of 5.4. However one can define, probably in many different ways, a canonical subscheme which gives the non-smooth locus in a general setting. We shall outline one here: Given an A -algebra B , define $I_{B|A} = \bigcap_E \text{Ann } D^1(B|A, E)$ where E runs through all B -modules of finite type. This ideal $I_{B|A}$ commutes with localizations by 3.13(c), and therefore, given a morphism $X \xrightarrow{f} Y$, we obtain the ideal sheaf $I_{X|Y}$. If f is a morphism of essentially finite type, then f is smooth at x if and only if $x \notin \text{Supp } O_X/O_X|_Y$ by 3.13(a). This notion is in an obvious way carried over to adic formal schemes of essentially

finite type and thus it is equally suitable for deformation theory.

6. Non-degenerate quadratic singularities

The objective in this section is to investigate the deformations of a k -scheme having non-degenerate quadratic singularities only. In view of 4.13, it becomes purely a local problem in an appropriate situation.

We go back to the notations of §4. Thus Λ is a fixed complete noetherian local ring with the residue field k , and $\underline{C}_\Lambda$ stands for the category of local artinian Λ -algebras with the same residue field k .

Let X be a k -scheme of finite type, and let x be a closed point of X . We recall that the point x is said to be a non-degenerate quadratic singularity rational over k if $\hat{O}_{X,x} = k[[x_1, \dots, x_n]]/(f)$ with $(\partial f/\partial x_1, \dots, \partial f/\partial x_n) = (x_1, \dots, x_n)$. One can always assume f to be a quadratic form. Indeed we have

Lemma 6.1. Let $f \in k[[x_1, \dots, x_n]]$. If $(\partial f/\partial x_1, \dots, \partial f/\partial x_n) = M (= (x_1, \dots, x_n))$, then one can find $x_1', \dots, x_n'$ with $x_i = x_i' \pmod{M^2}$ such that $f = q(x_1', \dots, x_n')$ where q is a quadratic form.

Proof: We do this by a successive approximation. Write $f = f_2 + h_3$ where f_2 is a quadratic form and $h_3 \in M^3$. Since $(\partial f/\partial x_1, \dots, \partial f/\partial x_n) = M$, there exists $\epsilon_1^{(1)}, \dots, \epsilon_n^{(1)}$ in M^2 such that $\sum \epsilon_i^{(1)} \partial f/\partial x_i \equiv h_3 \pmod{M^4}$. If we set $x_i^{(1)} = x_i - \epsilon_i^{(1)}$, then $f(x_1^{(1)}, \dots, x_n^{(1)}) = f(x_1, \dots, x_n) - \sum \epsilon_i^{(1)} \partial f/\partial x_i + \dots \equiv f_2(x_1, \dots, x_n) \pmod{M^4}$ i.e., $f(x_1^{(1)}, \dots, x_n^{(1)}) \equiv f_2(x_1, \dots, x_n) + h_4$ with $h_4 \in M^4$. Now $(\partial f/\partial x_1^{(1)}, \dots, \partial f/\partial x_n^{(1)}) = (\partial f/\partial x_1, \dots, \partial f/\partial x_n) = M$ and therefore we can choose

$\epsilon_1^{(2)}, \dots, \epsilon_n^{(2)}$ in M^3 such that $\sum \epsilon_i^{(2)} \partial f / \partial x_i^{(1)} \equiv h_4 \pmod{M^5}$. Then
 $f(x_1^{(1)} - \epsilon_1^{(2)}, \dots, x_n^{(1)} - \epsilon_n^{(2)}) = f(x_1^{(1)}, \dots, x_n^{(1)}) - \sum \epsilon_i^{(2)} \partial f / \partial x_i^{(1)} + \dots$
 $\equiv f_2(x_1, \dots, x_n) \pmod{M^5}$. Continuing this process we can find
 $\epsilon_1^{(1)}, \epsilon_1^{(2)}, \epsilon_1^{(3)}, \dots$ ($i = 1, 2, \dots, n$) with $\epsilon_i^{(v)} \in M^{v+1}$ such that if we set
 $x_i^{(v)} = x_i - (\epsilon_i^{(1)} + \dots + \epsilon_i^{(v)})$ then $f(x_1^{(v)}, \dots, x_n^{(v)}) \equiv f_2(x_1, \dots, x_n)$
 $\pmod{M^{v+3}}$. Thus if we set $\epsilon_i = \sum_v \epsilon_i^{(v)}$ then $f(x_1 - \epsilon_1, \dots, x_n - \epsilon_n) =$
 $= f_2(x_1, \dots, x_n)$ i.e., $f(x_1, \dots, x_n) = f_2(x_1 + \epsilon_1, \dots, x_n + \epsilon_n)$.

Lemma 6.2. Let A_0 be a local k -algebra, essentially of finite type such that

$$\hat{A}_0 \simeq k[[x_1, \dots, x_n]]/(f)$$

where f is a non-degenerate quadratic form, and let $(R, \mathcal{X})$ be a formal versal deformation of $X_0 = \text{Spec}(A_0)$. Then

- (1) $J_{\mathcal{X}}(\mathcal{X}|R) \rightarrow J_R(\mathcal{X}|R) \rightarrow \text{Spec}(\Lambda)$ are both isomorphisms.
- (2) $J_R(\mathcal{X}|R)$ is a principal ideal in R generated by a free formal generator of R over Λ i.e. $I_R(\mathcal{X}|R) = (t)$ and $R = \Lambda[[t]]$.
- (3) $\mathcal{X} \rightarrow \text{Spec}(\Lambda)$ is formally smooth.

Proof: We set $\mathcal{X} = \text{Spf}(\hat{A})$. By 4.10(b), $(R, \text{Spf}(\hat{A}))$ is a formal versal deformation of $\hat{X}_0 = \text{Spec}(\hat{A}_0)$ where $\hat{A}_0 = \varprojlim_v A \otimes_R R_v$.

Now $\hat{A}_0 \simeq k[[x_1, \dots, x_n]]/(f)$ and $(\partial f / \partial x_1, \dots, \partial f / \partial x_n) = (x_1, \dots, x_n)$ in $k[[x_1, \dots, x_n]]$. It follows readily that $\varprojlim_{X_0} (k[\epsilon]) \simeq k$ and, by 4.14, we get

$$R \simeq \Lambda[[t]] \text{ and } \hat{A} \simeq R[[x_1, \dots, x_n]]/(F - t)$$

where F is a quadratic polynomial in $\Lambda[x_1, \dots, x_n]$ ($\subset R[[x_1, \dots, x_n]]$) obtained from f by lifting the coefficients to Λ . Now

$$0 \rightarrow \hat{A} \xrightarrow{df} \hat{A}dx_1 \oplus \dots \oplus \hat{A}dx_n \xrightarrow{\hat{\pi}_{\hat{A}|R}} \hat{\pi}_{\hat{A}|R} \rightarrow 0$$

is exact where $df : \hat{A} \rightarrow \hat{A}dx_1 \oplus \dots \oplus \hat{A}dx_n$ is given by
 $1 \rightarrow (\partial f / \partial x_1)dx_1 + \dots + (\partial f / \partial x_n)dx_n$, and therefore
 $I_{\hat{A}}(\hat{A}|R) = (\partial f / \partial x_1, \dots, \partial f / \partial x_n) = (x_1, \dots, x_n)$, and hence
 $\hat{A}/I_{\hat{A}}(\hat{A}|R) = \hat{A}/(x_1, \dots, x_n) = R[[x_1, \dots, x_n]]/(x_1, \dots, x_n, F - t) \simeq \Lambda$
i.e., the canonical map $\Lambda \rightarrow \hat{A}/I_{\hat{A}}(\hat{A}|R)$ is an isomorphism. It follows that
 $\hat{A} \rightarrow A/I_A(A|R)$ is an isomorphism, and consequently $\Lambda \rightarrow R/I_R(A|R)$ is also an
isomorphism. This proves (1) and (2). On the other hand, (3) follows from
 $\hat{A} \simeq R[[x_1, \dots, x_n]]/(F - t) \simeq \Lambda[[t, x_1, \dots, x_n]]/(F - t) \simeq \Lambda[[x_1, \dots, x_n]]$ as
 Λ -algebras.

Corollary 6.3. Let X_0 be a k -scheme of finite type, and let x in X_0 be a non-degenerate quadratic singularity rational over k .

(1) Let $(R, \mathcal{I})$, $(R, \mathcal{U})$ be any two deformations of X_0 where $R \in \text{ob}(\hat{\mathcal{C}}_\Lambda)$. Assume that $k = k^2$. Then $\mathcal{I}_{(\mathcal{I})} \simeq \mathcal{U}_{(\mathcal{U})}$ as R -adic formal schemes if and only if $I_R(\mathcal{I}_{(x)}|R) = I_R(\mathcal{U}_{(x)}|R)$.

(2) Let $R \in \text{ob}(\hat{\mathcal{C}}_\Lambda)$ be a regular local ring, and let $(R, \mathcal{I})$ be a deformation of X_0 . Then $\mathcal{O}_{\mathcal{I}, x}$ is a regular local ring if and only if $I_R(\mathcal{I}_{(x)}|R) \not\subset \mathfrak{m}_R^2$ i.e., if and only if $I_R(\mathcal{I}_{(x)}|R)$ is a regular divisor in $\text{Spf}(R)$.

Proof: Let (R_0, Z) be a formal versal deformation of $\text{Spec}(\hat{\mathcal{O}}_{X_0, x})$. Since x in X_0 is a non-degenerate quadratic singularity rational over k , we may assume that $R_0 \simeq \Lambda[[t]]$ and $Z \simeq \text{Spf}(R_0[[x_1, \dots, x_n]]/(F - t))$ as in 6.2.

(1) $(R, \mathcal{I}_{(\mathcal{I})})$, $(R, \mathcal{U}_{(\mathcal{U})})$ are deformations of $\text{Spec}(\hat{\mathcal{O}}_{X_0, x})$, and therefore they are induced from (R_0, Z) by continuous local Λ -algebra maps $\varphi, \psi : R_0 \rightarrow R$ respectively, i.e., if we set $a = \varphi(t)$, $b = \psi(t)$, then

$$\mathcal{I}_{(\mathcal{I})} \simeq \text{Spf}(R[[x_1, \dots, x_n]]/(R - a)),$$

$$\mathcal{U}_{(\mathcal{U})} \simeq \text{Spf}(R[[x_1, \dots, x_n]]/(F - b))$$

as R -adic formal schemes. Assume that $I_R(\mathcal{I}_{(x)}|R) = I_R(\mathcal{U}_{(x)}|R)$ i.e.,

(a) = (b) in R . Then $a = ab$ for some unit u in R . Since $k = k^2$ and R is complete, we can write $u = v^2$ for some unit v in R , i.e., $a = v^2 b$. Then $(F - b) = (v^2 F - a)$, so that $X_i \rightarrow vX_i$ establishes an R -algebra isomorphism $R[[X_1, \dots, X_n]]/(F - a) \xrightarrow{\sim} R[[X_1, \dots, X_n]]/(F - b)$ since F is a quadratic form. The other direction is trivial.

(2) There exists a continuous local Λ -algebra map $\varphi : R_0 = \Lambda[[t]] \rightarrow R$ such that $\mathcal{I}_{(\hat{x})} \simeq \text{Spf}(R[[X_1, \dots, X_n]]/(F - \varphi(t)))$. We note that $I_R(\mathcal{I}_{(x)})|R = I_R(\mathcal{I}_{(\hat{x})})|R = (\varphi(t))$, and that $R[[X_1, \dots, X_n]]/(F - \varphi(t)) = \hat{\mathcal{O}}_{\mathcal{I}, x}$ is regular if and only if $\mathcal{O}_{\mathcal{I}, x}$ is regular. Now R is regular and hence $R[[X_1, \dots, X_n]]$ is also a regular local ring. Thus if we set M to be the maximal ideal of $R[[X_1, \dots, X_n]]$, then $R[[X_1, \dots, X_n]]/(F - \varphi(t))$ is regular if and only if $F - \varphi(t)$ is not in M^2 , i.e. if and only if $\varphi(t) \notin \mathfrak{m}_R^2$, i.e., if and only if $I_R(\mathcal{I}_{(x)})|R \notin \mathfrak{m}_R^2$.

Theorem 6.4. Let X_0 be a k -scheme of finite type with isolated non-smooth points only, and let $(R, \mathcal{I})$ be a formal versal deformation of X_0 . Denote by $\{z_1, \dots, z_m\}$ the set of non-smooth points in X_0 . Assume that $H^2(X_0, D_{X_0}^0) = 0$, and all z_i 's are non-degenerate quadratic singularity rational over k . Then

(1) R is formally smooth over Λ i.e., R is a formal power-series ring over Λ ;

(2) $J_{\mathcal{I}}(\mathcal{I}|R) = \coprod_{1 \leq i \leq m} J_{\mathcal{I}_{(z_i)}}(\mathcal{I}_{(z_i)}|R)$ (disjoint union), and

$J_{\mathcal{I}_{(z_i)}}(\mathcal{I}_{(z_i)}|R) \xrightarrow{\sim} J_R(\mathcal{I}_{(z_i)}|R)$ is an isomorphism;

(3) $I_R(\mathcal{I}_{(z_i)}|R)$ is a principal ideal and in turn so is

$I_R(\mathcal{I}|R) = \prod_{1 \leq i \leq m} I_R(\mathcal{I}_{(z_i)}|R)$. Furthermore if we set $I_R(\mathcal{I}_{(z_i)}|R) = (t_i)$, then

$t_1, \dots, t_m$ can be extended to a system of free formal generators of R over Λ (as formal power-series ring over Λ).

(4) $\mathcal{I} \rightarrow \text{Spec}(\Lambda)$ is smooth, i.e. $J_{\mathcal{I}}(\mathcal{I}|\Lambda) = \varphi$.

(5) If X_0 is a complete irreducible curve, then $\dim M_{X_0} = 3g - 3 + a$ where $g = \dim_K H^1(X_0, \mathcal{O}_{X_0})$ and $a = \dim_K H^0(X_0, \mathcal{D}_{X_0}^0)$.

Proof: (1) follows from (2) and (3): The first statement of (2) follows from 5.7(2). Now let $(R_1, \mathcal{U}_1)$ be a formal versal deformation of $\text{Spec}(\hat{\mathcal{O}}_{X_0, z_1})$. Since $(R, \mathcal{X}_{(z_1)})$ is a deformation of $\text{Spec}(\hat{\mathcal{O}}_{X_0, z_1})$, there exists a continuous local Λ -algebra maps $\varphi_1 = R_1 \rightarrow R$ with a cartesian diagram

$$\begin{array}{ccc} \mathcal{X}_{(z_1)} & \xrightarrow{\quad} & \mathcal{U}_1 \\ \downarrow & & \downarrow \\ \text{Spf}(R) & \xrightarrow{\text{Spf}(\varphi_1)} & \text{Spf}(R_1) \end{array}$$

Therefore we have

$$\begin{aligned} J_{\mathcal{X}_{(z_1)}}(\mathcal{X}_{(z_1)}|R) &= J_{\mathcal{X}_{(z_1)}}(\mathcal{X}_{(z_1)}|R) \simeq J_{\mathcal{U}_1}(\mathcal{U}_1|R_1) \times_{\text{Spf}(R_1)} \text{Spf}(R) \\ J_R(\mathcal{X}_{(z_1)}|R) &= J_R(\mathcal{X}_{(z_1)}|R) \simeq J_{R_1}(\mathcal{U}_1|R_1) \times_{\text{Spf}(R_1)} \text{Spf}(R). \end{aligned}$$

Since $J_{\mathcal{U}_1}(\mathcal{U}_1|R_1) \rightarrow J_{R_1}(\mathcal{U}_1|R_1)$ is an isomorphism by 6.2(1), we have that the canonical morphism $J_{\mathcal{X}_{(z_1)}}(\mathcal{X}_{(z_1)}|R) \rightarrow J_R(\mathcal{X}_{(z_1)}|R)$ is an isomorphism. Now

$I_{R_1}(\mathcal{U}_1|R_1) = (u_1) = (u_1)$ with $R_1 = \Lambda[[u_1]]$ by 6.2(2), and therefore

$$I_R(\mathcal{X}_{(z_1)}|R) = I_R(\mathcal{X}_{(z_1)}|R) = (t_1)$$

where $t_1 = \varphi_1(u_1)$, and therefore $I_R(\mathcal{X}|R) = \prod_{1 \leq i \leq m} I_R(\mathcal{X}_{(z_i)}|R) = (\prod_{i=1}^n t_i)$.

Now $M_{X_0} \rightarrow M_{X_0, (z_1)} \times \dots \times M_{X_0, (z_m)}$ is smooth by and therefore

$R_1 \hat{\otimes}_{\Lambda} \dots \hat{\otimes}_{\Lambda} R_m \rightarrow R$ is formally smooth by 4.13, which implies that $t_1, \dots, t_m$ is extended to a system of free formal generators of R over Λ .

(4) The projection morphism $\mathcal{X}_{(z_1)} \cong \mathcal{U}_1 \times_{\text{Spf}(R_1)} \text{Spf}(R) \rightarrow \mathcal{U}_1$ is formally smooth since $\text{Spf}(R) \xrightarrow{\text{Spf}(\varphi_1)} \text{Spf}(R_1)$ is formally smooth. On the other hand, $\mathcal{U}_1 \rightarrow \text{Spec}(\Lambda)$ is formally smooth by 6.2(3) and therefore $\mathcal{X}_{(z_1)} \rightarrow \text{Spec}(\Lambda)$ is formally smooth. Consequently $\mathcal{X} \rightarrow \text{Spec}(\Lambda)$ is formally smooth since $\{z_1, \dots, z_m\}$ is the set of non-smooth points of X_0 .

(5) Since R is formally smooth over Λ and

$$0 \rightarrow H^1(X_0, \underline{D}_{X_0}^0) \rightarrow [M_{X_0}(k[\epsilon])] \rightarrow H^0(X_0, \underline{D}_{X_0}^1) \rightarrow 0$$

is exact by 4.4 and 4.11, we have

$$\begin{aligned} \dim M_{X_0} &= \dim k \otimes R = \dim_k H^1(X_0, \underline{D}_{X_0}^0) + \dim H^0(X_0, \underline{D}_{X_0}^1) \\ &= \dim_k H^1(X_0, \underline{D}_{X_0}^0) + m. \end{aligned}$$

Then it suffices to show that

$$\dim_k H^1(X_0, \underline{D}_{X_0}^0) + m = 3g - 3 + \dim_k H^0(X_0, \underline{D}_{X_0}^0)$$

$$\text{i.e., } \chi(\underline{D}_{X_0}^0) = 3 - 3g + m$$

where we set $\chi(\underline{E}) = \dim_k H^0(X_0, \underline{E}) - \dim_k H^1(X_0, \underline{E})$ for any coherent X_0 -module $\underline{E}$. Since $\chi(\underline{D}_{X_0}^0)$ does not change when one replaces the base field k by its

algebraic closure, we may assume that k is algebraically closed. Set $A = \underline{O}_{X_0, z}$, where z is a non-smooth point of X_0 , and let A' be its normalization.

By hypothesis, we have $A \cong k[[x, y]]/(f)$ where f is a non-degenerate quadratic form. Since k is algebraically closed, we may assume that $f = xy$, i.e.,

$A \cong k[[x, y]]/(xy)$. From this we conclude immediately that $\ell(A'/A) = 1$ and

$\text{Der}_k(A, \underline{m}_A) = \text{Der}_k(A, A)$. Let C be the conductor of A in A' . Since A

is a complete-intersection we must have that $\ell(A/C) = \ell(A'/A) = 1$ so that

$C = \underline{m}_A$. We thus obtain that $A' = \{a' \in K \mid a' \underline{m}_A \subset \underline{m}_A\}$ where K is the fraction field of A . We now claim that $\text{Der}_k(A, A) \subset \text{Der}_k(A', A')$ as A -modules contained in $\text{Der}_k(K, K)$. Indeed, let $X \in \text{Der}_k(A, A)$. Then for any $a' \in A'$ and

$x \in \underline{m}_A$ we have $xa' \in \underline{m}_A$ so that $X(xa') \in \underline{m}_A$ and $a'X(x) \in \underline{m}_A$ since $\text{Der}_k(A, \underline{m}_A) = \text{Der}_k(A, A)$. Consequently $X(xa') = a'X(x) + xX(a')$ entails that $xX(a') \in \underline{m}_A$ for all $x \in \underline{m}_A$, and therefore $X(a') \in A'$. Furthermore, the exact sequence

$$0 \rightarrow \text{Der}_k(A, A) \rightarrow \text{Der}_k(A', A') \rightarrow A'/\underline{m}_A A' \rightarrow 0$$

entails the exact sequence

$$0 \rightarrow \text{Der}_k(A, A) \rightarrow \text{Der}_k(A', A') \rightarrow A'/C \rightarrow 0$$

Therefore if we set X'_0 to be the normalization of X_0 , then

$$0 \rightarrow \underline{D}_{X_0}^0 \rightarrow \underline{D}_{X'_0}^0 \rightarrow \underline{O}_{X'_0}/C \rightarrow 0$$

is exact. Since X_0 is locally a complete-intersection, we have

$\ell(\underline{O}_{X'_0}/C) = 2\ell(\underline{O}_{X'_0}/\underline{O}_{X_0}) = 2m$, whereas $\chi(\underline{D}_{X'_0}^0) = 3 - 3(g-m)$ by Riemann-Roch theorem. Consequently

$$\chi(\underline{D}_{X_0}^0) = \chi(\underline{D}_{X'_0}^0) - \ell(\underline{O}_{X'_0}/C) = 3 - 3(g-m) - 2m = 3 - 3g + m.$$

This completes our proof.

Remark 6.5. In the formula 6.4(5), the number $a = H^0(X_0, \underline{D}_{X_0}^0)$ can be easily computed in terms of Riemann-Roch theorem. Thus

$$a = \begin{cases} 0 & \text{if } g \geq 2 \\ 1 & \text{if } g = 1 \\ 3 & \text{if } g = 0 \end{cases}$$

Finally we add a remark on characteristic 2. If the characteristic of the base field k is 2, there is no non-degenerate quadratic form on n variables if n is odd. In other words there does not exist a non-degenerate quadratic singularity of even dimension if $\text{char } k = 2$. In this case one may consider a quadratic form of maximum non-degeneracy:

Definition 6.6. Let X be a k -scheme of finite type. A closed point $z \in X$ is called an ordinary quadratic singularity rational over k if the following holds:

(i) If either $\text{char } k \neq 2$ or $\text{char } k = 2$ and $\dim \mathcal{O}_{X,z}$ is odd, then a is a non-degenerate quadratic singularity rational over k .

(ii) If $\text{char } k = 2$ and $\dim \mathcal{O}_{X,z}$ is even, then $\bar{k} \otimes_k \hat{\mathcal{O}}_{X,z} \simeq \bar{k}[[X_0, X_1, \dots, X_{2n}]]/(f)$ with

$$f = X_0^2 + X_1X_2 + X_3X_4 + \dots + X_{2n-1}X_{2n}$$

where $\bar{k}$ is the algebraic closure of k .

Lemma 6.7. Let k be a field of characteristic 2 and set

$$A_0 = k[[X_0, X_1, \dots, X_{2n}]]/(f) \text{ where}$$

$$f = X_0^2 + X_1X_2 + X_3X_4 + \dots + X_{2n-1}X_{2n},$$

and let $(R, \mathcal{I})$ be a formal versal deformation of $X_0 = \text{Spec}(A_0)$. Then

$R = \Lambda[[t, t_1]]$, where t, t_1 are free formal generators of R over Λ , such that $I_R(\mathcal{I}|R) = (t^2)$, and $\mathcal{I} \rightarrow \text{Spec}(\Lambda)$ is formally smooth.

Proof: We set $\mathcal{I} = \text{Spf}(A)$. Since

$$(f, \frac{\partial f}{\partial X_0}, \dots, \frac{\partial f}{\partial X_{2n}}) = (X_0^2, X_1, \dots, X_{2n}),$$

it follows that $D^1(A_0|k, A_0) \simeq k[X_0]/(X_0^2)$ and therefore $[M_{X_0}(k[\epsilon])]$ is 2-dimensional vector space over k . Furthermore, it follows from 4.14 that

$$R \simeq \Lambda[[t, t_1]] \text{ and } A \simeq R[[X_0, X_1, \dots, X_{2n}]]/(F)$$

where $F = f + t_1 + tX_0$. Now the exact sequence

$$0 \rightarrow A \xrightarrow{dF} \text{Ad}X_0 \oplus \dots \oplus \text{Ad}X_{2n} \rightarrow \hat{\Omega}_A|_R \rightarrow 0$$

where

$$dF = 1 \rightarrow \sum_{0 \leq i \leq 2n} \frac{\partial f}{\partial X_i} dX_i + dX_0,$$

entails that

$$I_A(A|R) = (t, x_1, x_2, \dots, x_{2n}) = (x_0^2 + t_1, t, x_1, \dots, x_{2n})$$

so that

$$A/I_A(A|R) \simeq A/(x_0^2 + t_1, t, x_1, \dots, x_{2n}) \simeq \Lambda[[t_1, x_0]]/(x_0^2 - t_1) .$$

Since $\Lambda[[t_1, x_0]]/(x_0^2 - t_1) \simeq R/(t) \oplus R/(t)$ as R -modules, it follows that

$I_R(A|R) = (t^2)$, which proves our first assertion. On the other hand,

$$\begin{aligned} A &\simeq \Lambda[[t, t_1, x_0, x_1, \dots, x_{2n}]]/(x_0^2 + x_1x_2 + \dots + x_{2n-1}x_{2n} + tx_0 + t_1) \\ &\simeq \Lambda[[t, x_0, x_1, \dots, x_{2n}]] \end{aligned}$$

and therefore A is formally smooth over Λ .

The above statement 6.7 yields a similar result as 6.4, proof of which we leave to the readers.

Corollary 6.8. Let X_0 be a k -scheme of finite type with isolated non-smooth points only such that $H^2(X_0, \underline{D}_{X_0}^0) = 0$, and let $(R, \mathcal{I})$ be a formal versal deformation of X_0 . Denote by $\{z_1, \dots, z_m\}$ the set of non-smooth points in X_0 , and assume that $\text{char } k = 2$, $\dim X_0$ is even, and all z_i 's are ordinary quadratic singularities. Then

(1) R is formally smooth over Λ

(2) $I_R(\mathcal{I}_{(z_i)}|R)$ is a principal ideal and in turn so is

$I_R(\mathcal{I}|R) = \prod_{1 \leq i \leq m} I_R(\mathcal{I}_{(z_i)}|R)$. Furthermore, $I_R(\mathcal{I}_{(z_i)}|R) = (t_i^2)$ and

$t_1, \dots, t_m$ can be extended to a system of free formal generators of R over Λ .

(3) $\mathcal{I} \rightarrow \text{Spec}(\Lambda)$ is formally smooth

(4) If k is algebraically closed and X_0 is a complete irreducible curve, then $\dim M_{X_0} = 3g - 3 + a$ where $g = \dim_k H^1(X_0, \mathcal{O}_{X_0})$ and $a = \dim_k H^0(X_0, \underline{D}_{X_0}^0)$.